

Worksheet — Friction on an Incline Simulation

Simulation 03 • use with *Simulation - Friction on an incline.html* • goes with the Forces investigation (friction on a ramp)

Name: _____ Class: _____ Date: _____

How to use this worksheet

Open **Simulation - Friction on an incline.html** in a browser (it works offline). ▲ **Tilt** raises the ramp until the block slips; ▶ **Release** lets the block go at the angle on the slider. The **Tasks** tab has the same three parts as this sheet; write your **explanations** here.

Part 1 before the lab (≈ 15 min) • **Part 2** during the lab (≈ 10 min) • **Part 3** homework (≈ 30 min).

Use $g = 9.81 \text{ N kg}^{-1}$. Sliding uses Newton's 2nd law (Blocks 3–4).

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. The ramp tilts and the block stays at rest. What happens to the friction force?			
2. What happens to the normal force as the ramp tilts?			
3. You double the block's mass. What happens to the slip angle?			
4. Which surfaces give the smallest slip angle: rubber, wood/bench, ice or sandpaper?			
5. After it slips (ramp angle fixed), does the block slide at constant speed, speed up or stop?			

6. Draw the free-body diagram of the block at rest on a 20° ramp. Show the weight components $mg \sin \theta$ and $mg \cos \theta$ as dashed arrows.

Free-body diagram

7. Show that the block slips when $\tan \theta = \mu_s$. Then calculate the slip angle for wood on the bench ($\mu_s = 0.41$).

Working

$\theta_{\text{slip}} = \underline{\hspace{2cm}}^\circ$ Sim check: $\underline{\hspace{2cm}}^\circ$

Part 2 — During the lab: Forces investigation

Investigation block (0.150 kg) on the wooden board. Fill in the sim values and your group's values.

Quantity	Simulation	Our measurement / calculation
slip angle $\theta_{\text{slip}} / ^\circ$		
$\mu_s = \tan \theta_{\text{slip}}$		
largest static friction at 20° , $\mu_s mg \cos 20^\circ / \text{N}$		
friction acting at 10° , $mg \sin 10^\circ / \text{N}$		
acceleration at 60° , $a / \text{m s}^{-2}$		7.8 (given)
$\mu_d = (g \sin \theta - a) / (g \cos \theta)$		

8. Give **one** reason why your measured slip angle differs from the simulation's.

9. At 10° the friction is much smaller than $\mu_s F_N$. Explain why.

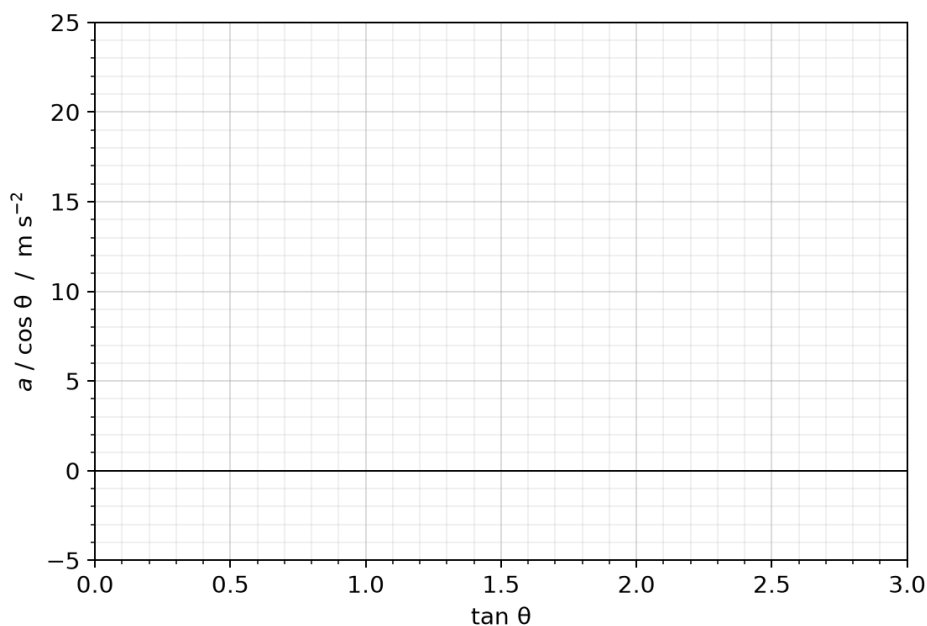
Part 3 — Homework: find μ_d from a graph

Wood on bench, $m = 0.500$ kg. Release the block at each angle and record a .

$\theta / ^\circ$	30	40	50	60	70
$\tan \theta$					
$a / \text{m s}^{-2}$					
$\cos \theta$					
$a / \cos \theta / \text{m s}^{-2}$					

10. Show that $a = g(\sin \theta - \mu_d \cos \theta)$ can be written as $a/\cos \theta = g \tan \theta - \mu_d g$. What should the gradient and the intercept of a graph of $a/\cos \theta$ against $\tan \theta$ be?

11. Plot $a/\cos \theta$ against $\tan \theta$ and draw the best-fit line.



12. Gradient = _____ m s^{-2} Intercept = _____ $\text{m s}^{-2} \Rightarrow \mu_d =$ _____

13. Calculate the acceleration of the block sliding down the bench at 35° ($\mu_d = 0.30$) and the time it takes to slide 0.80 m from rest.

Working

14. The 0.500 kg block is released on the bench at 15° ($\mu_s = 0.41$). Draw its free-body diagram. Does it move? Calculate the friction force acting and compare it with $\mu_s F_N$.

Free-body diagram

Working

15. A car is parked on a steep hill. On a dry road it stays put; on ice it slides down. Explain why, using $\tan \theta$ and μ_s . Does a heavier car slide more easily?
